Active Inference – A Synthetic Overview

Extended Abstract

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A thing that persists must, in effect, hold a model of the world that keeps it intact. **Active Inference** turns that observation into mechanics: a system maintains itself by minimizing *variational free energy*, a single quantity that scores how badly its internal expectations fit incoming sensations (Friston 2010; Parr et al. 2022). Perception lowers free energy by updating beliefs; action lowers it by changing the world so the beliefs come true. One currency, two directions — inference and control — which is why the same template reappears from single cells to brains to institutions.

The bound is what makes this *applicable* rather than metaphorical. Variational free energy is an upper bound on **surprise**, the negative log evidence $-\ln p(s)$ a system assigns to its own sensory states. Because the bound is computable from a system's own beliefs, minimizing it *is* approximate Bayesian inference carried out by the physical dynamics of the thing itself. The accounting is organized by a **Markov blanket**: a statistical boundary splitting *external* states η from *internal* states μ, coupled only through *sensory* and *active* blanket states. Internal states come to parameterize beliefs about the external world they can never touch directly — the core claim of **Bayesian mechanics**, a physics written in terms of, and by, those beliefs (Da Costa et al. 2021; Ramstead et al. 2023).

The figure below shows the move this overview is built around, without a single equation of derivation. Both panels live on *the same* free-energy landscape — a Gaussian bowl whose floor is the system's expected state. At **equilibrium** (panel a) the flow is pure descent, $f = -\nabla F$: every trajectory falls straight to the mode and stops. Detailed balance holds, there is no net current, and — taken literally — nothing further happens. A rock is at equilibrium; so, eventually, is a system that has stopped doing anything. Living systems occupy panel (b). There the flow gains a *solenoidal* term, $f = (Q - \Gamma)\nabla F$, that runs along the contours of F rather than down them. Free energy is still minimized — the system still tracks its niche — yet probability mass **circulates** indefinitely in a **non-equilibrium steady state**: heartbeats, foraging, perception that never settles. The gradient part is the Free Energy Principle's "climbing down surprise"; the solenoidal part is what keeps the system alive while doing so (Friston et al. 2023).

Read the two panels together and the headline of Active Inference is visible at a glance: the *same* imperative — minimize free energy — produces inert relaxation in one regime and open-ended, self-sustaining behavior in the other. The rest of this overview unpacks that picture in the spirit of *show, not tell*: each subsequent section adds one runnable, minimal demonstration rather than another layer of formalism.

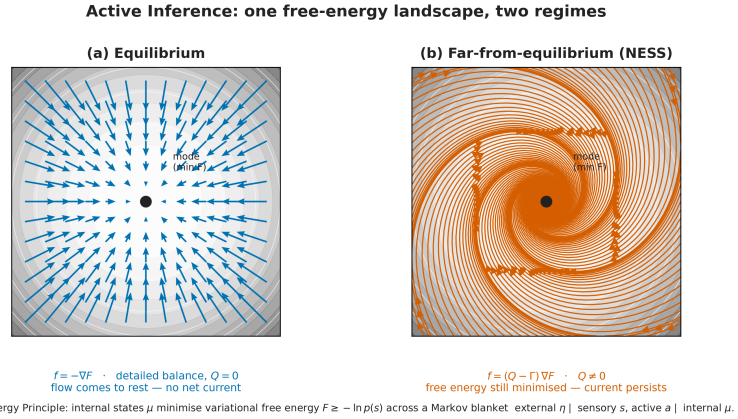


Figure 1: **One landscape, two regimes.** Same Gaussian free-energy bowl (grey contours; black dot = the mode, where F is least). **(a) Equilibrium:** pure gradient descent $f = -\nabla F$ (blue) — flow falls to the mode and rests, no net current. **(b) Far-from-equilibrium (NESS):** an antisymmetric *solenoidal* Q adds a conservative current $f = (Q - \Gamma)\nabla F$ (orange) along the level sets of F; free energy is still minimized, but probability mass circulates without end. FEP is the descent; Bayesian mechanics adds the circulation.

References

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